

DARREN GREGORY THE 9TH DELEGATE

COURTESY OF DARREN GREGORY

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BACHELOR OF EDUCATION AND BACHELOR OF SCIENCE

SBT 120

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CELLULAR BASIS OF LIFE

LESSON 1: THE CELL THEORY

- INTRODUCTION
- Every living organism is composed of one or more cells.
- Robert Hooke, in 1665, was the first person to use the term cell.
- He used this term in describing structures he observed in thin slices of plant tissue of cork using a microscope; he named them from the Latin word *cella meaning* a room where grain or fruit was stored.

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- This is because a cell or room has three dimensions, width, length and height and has contents (the idea of a store-room).
- A **cell** can be defined as the smallest organized unit of any living form which is capable of prolonged independent existence and replacement of substances that compose it.
- This definition includes plant and animal cells, bacteria and other micro-organisms.

characteristics of living things

The living organism possesses characteristics such as;

- They are able to acquire and utilize energy,
- Are able to produce more of them-selves,
- Possess a genetic program and a means to use it,
- Carry out a variety of chemical reactions and mechanical activities
- Are able to respond to stimuli and are capable of self-regulation.

LESSON 2: ORIGIN OF CELLS

- It is important to note that all cells originate from other cells hence implying that cells and living things initially originated from a cell, a cell-like or pre-cell structure from which all living organisms have evolved or changed from.
- The make-up of the cell includes fundamental chemical elements that are on the other hand made up of atoms.
- . Various theories have been put across to provide a possible description of how life came into existence.

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- Theories covered in this lesson include mainly the:

- 1. Scientific theory,**

- 2. the Spontaneous generation theory.**

- 3. the Creation theory**
- 1. The scientific theory of the origin of life**

The most supported hypothesis (Kant- Laplace hypothesis) about the origin of the solar system is that it began as a vast revolving cloud of gas and dust in space. This dust was made up of elements such as oxygen, hydrogen, carbon, nitrogen, silicon, iron, nickel and aluminium.

. As the heavier elements settled at the bottom, the lighter elements escaped to the cooler atmospheric layer where some formed compounds such as methane, Ammonia, Cyanogen, Steam and Carbon dioxide.

The earth was thus made of the core, mantle and a thin layer of crust. From the formed compounds such as cyanogen and methane, polymerization occurred to build more complex polymers such as amino acids, fatty acids, sugars, and nucleotides.

From the organic compounds, cells were formed in an environment containing organic compounds- an organic soup that nourished the organisms.

All organisms were initially anaerobic

- 2. Spontaneous generation theory of origin of life**

Spontaneous generation of life as originally proposed is the oldest theory, it postulated that life arise from nonliving matter. In its old original form the theory proposes that organisms (e.g. insects, mice) are generated spontaneously from mud or decaying organic matter.

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The original theory was developed by early Greeks, it is explained in Aristotle's writings (384 - 322 BC).

Aristotle believed that mosquitoes and fleas arose from decaying matter, tadpoles, worms and other small organisms arose from mud, flies from rotting meat. Aristotle was a respected intellectual of his time but he has been shown not to have been a careful observer.

He wrote for example that a man has more teeth than a woman, this is obviously wrong.

For a long time these ideas were not challenged until *Redi* (1626 - 1697 AD) an Italian physician attacked this theory and showed that meat covered with muslin cloth did not have maggots appearing from it.

3. Creation theories of origin of life

Many religions teach that a super human being God made the earth and everything in it as it is or in some basic forms. Ancient Babylonians taught that the god *Marduk* took goddess *Tiamat* and split her like a shellfish into two parts, half became the earth and the other half the sky.

. Blood vessels of another god made mankind and all living things. Ancient Egyptians thought the god Ra created mankind from his tears.

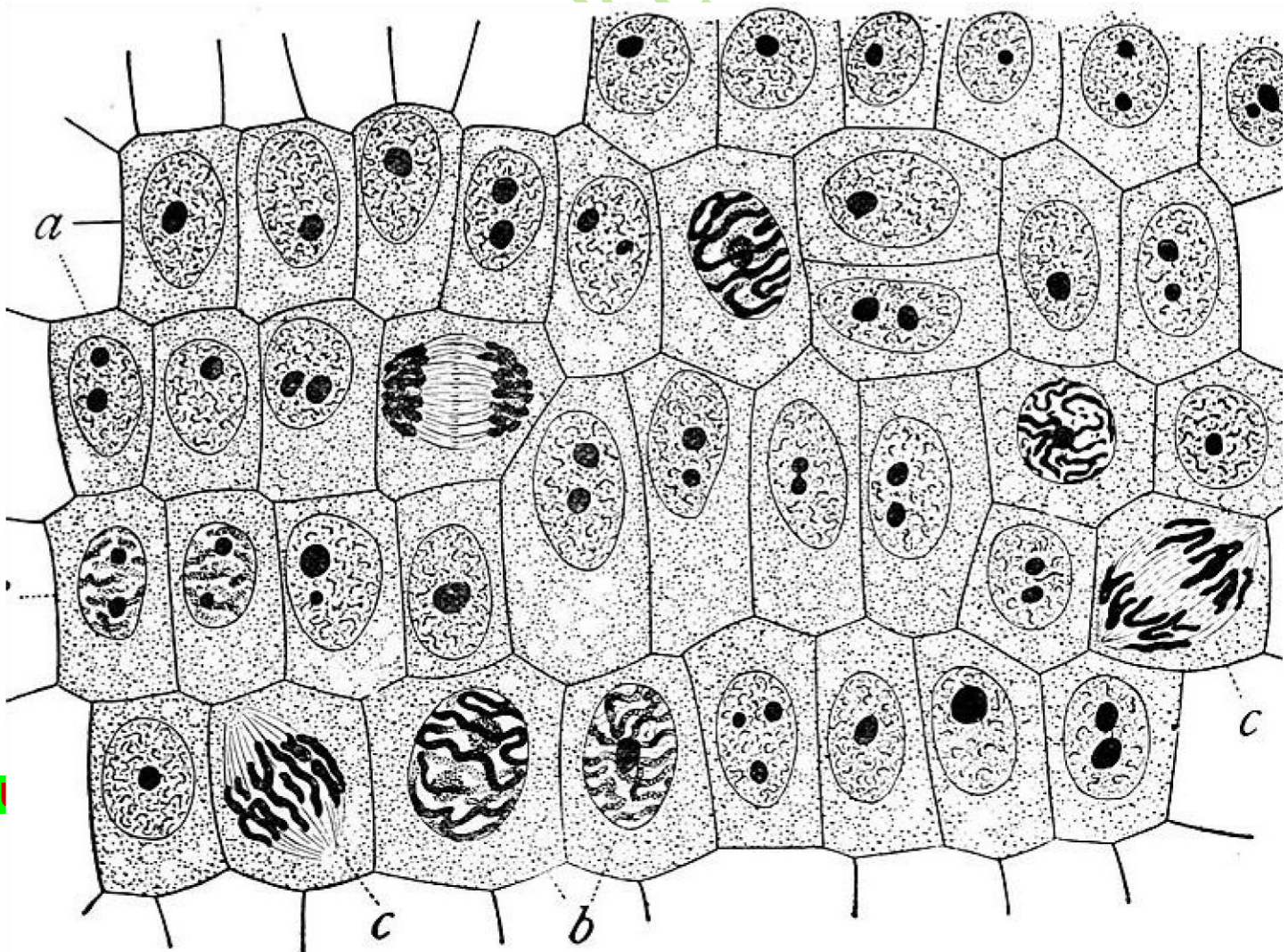
The Christian Bible in the book of Genesis says that God created heavens and the earth in seven days, higher animals were created according to their kind. All cultures have their own myths about the origin of life, the earth and the universe, e.g. Hindu Creation myth:

Initially a turtle carried 21 words consisting of the physical and spiritual universe.

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LESSON 3: THE CELL ORGANISATION

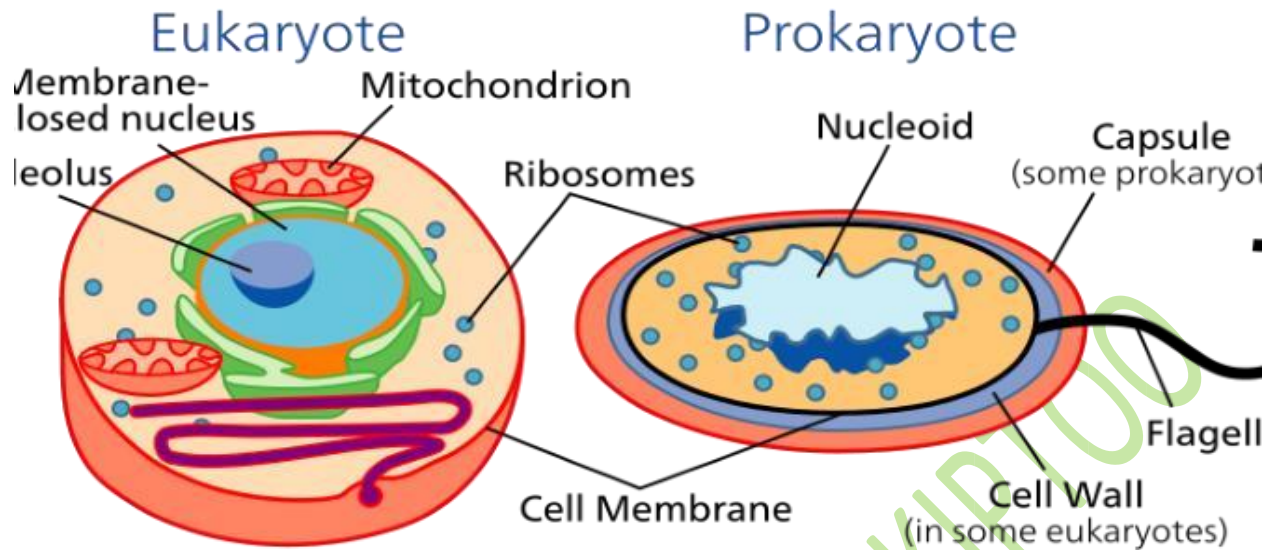
- Organisms can be classified into *acellular* (e.g. viruses), and those with cellular organization. In 1937, Chatton proposed the terms *procariotique* and *eucariotique* for non-nucleated and nucleated cellular organisms respectively.
- In addition, the lesson will focus in explaining the various similarities and differences between the animal and plant cells.



Classification of organisms by cell structure

- Organisms composed of more than one cell are said to be multicellular, an organism with a single cell is unicellular or acellular.
- Non-nucleated and nucleated cellular organisms are classified as prokaryotes and eukaryotes respectively.
- The eukaryotes (*Animalia, Fungi, Plantae and protista*) differ from prokaryotes (bacteria) in terms of their form, size, genetic material, protein, protein synthesis, cell wall, flagella, respiration, photosynthesis and in other ways as will be discussed in this section.

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Eukaryote vs Prokaryote

View diagram with the similarities and the differences between eukaryotic and prokaryotic cells

Difference between prokaryotic and eukaryotic cells

- Characteristics of prokaryotic cells include:

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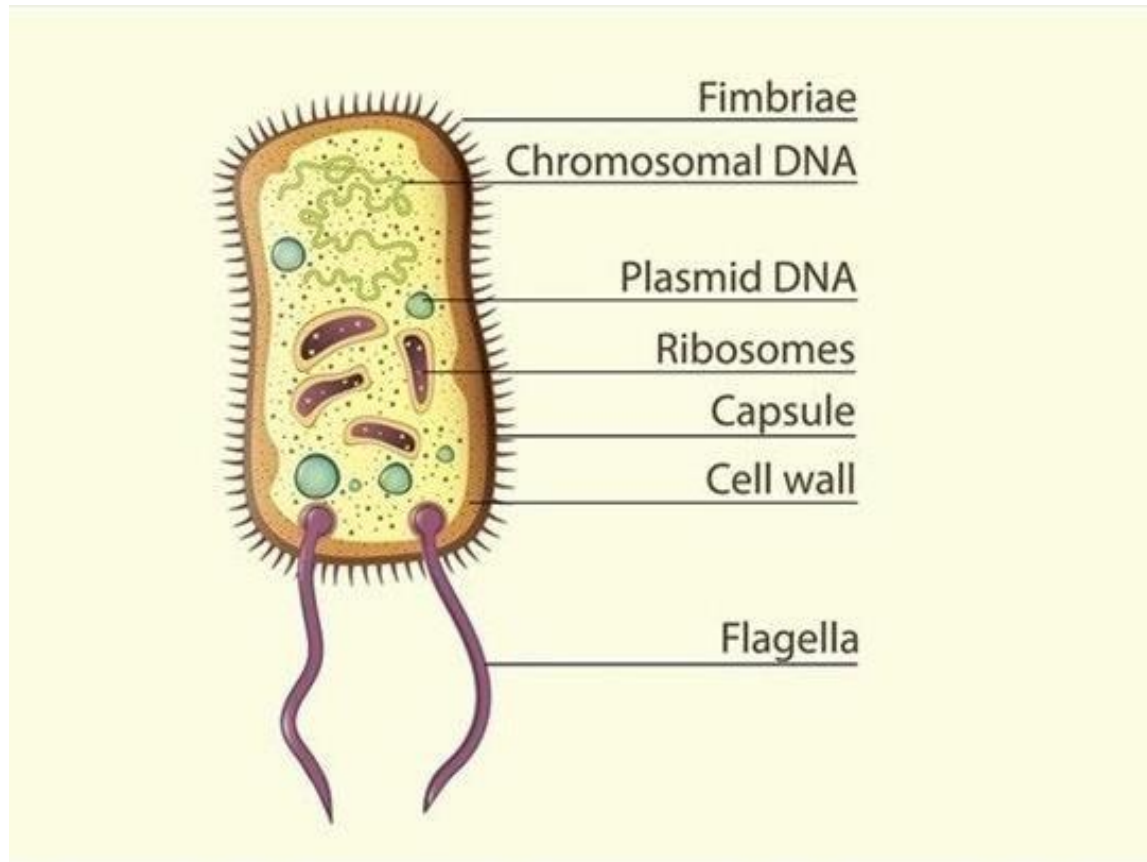
- i. They have no nucleus. Instead, some such as bacteria have a region within the cell where the genetic material is freely suspended. This region is called the nucleoid.
- ii. They are single-celled microorganisms. For example archaea, bacteria and cyanobacteria.
- iii. The cell size ranges from 0.1 to 0.5 μm in diameter. iv. iv.

The hereditary material can either be DNA or RNA.

- v. v. Prokaryotes generally reproduce by binary fission, a form of asexual reproduction. They are also known to use conjugation which is often seen as the prokaryotic equivalent of sexual production.

Structure of a prokaryotic cell

Structure of a prokaryotic cell



On the other hand, there are eukaryotic cells, and their characteristics include:

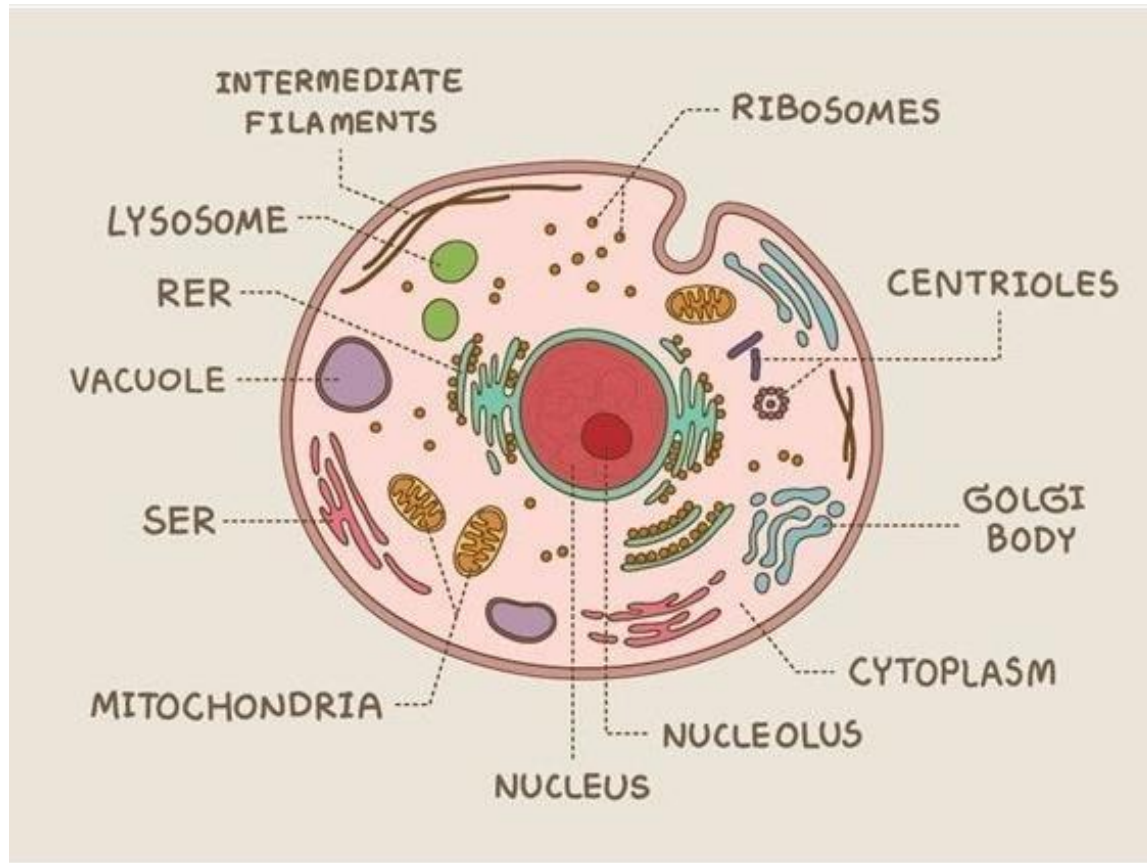
- i. They are characterized by a true nucleus.
- ii. The size of the cells ranges between 10-100 μm .

- iii. They reproduce sexually as well as asexually. iv. The broad category involves plants, fungi, protozoans and animals.
- iv. The plasma membrane is responsible for monitoring the transport of nutrients and electrolytes in and out of the cells. It is also responsible for cell to cell communication.

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Structure of a prokaryotic cell

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Structure of a Eukaryotic cell

General structure of the cell

- Cells differ in terms of their sizes and shapes. The size of a cell can be large as an egg cell cells (e.g. ostrich egg) which may measure several centimetre in diameter or 500 ml or more in volume down

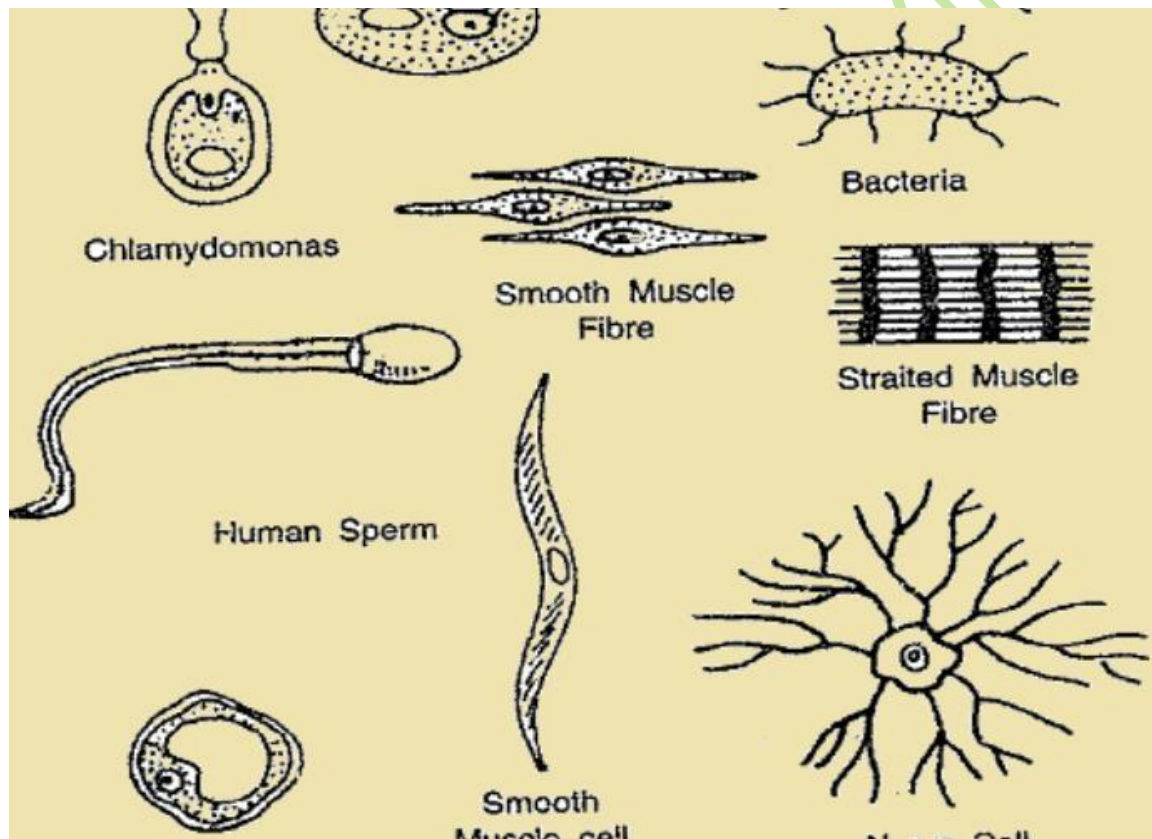
to minute bacterial cells of about 10^{13} ml or as small as 0.0001 of a millimetre.

- The shape of cells is highly variable ranging from regular shapes to irregular shapes, spherical to columnar and often some cells do not have definite shapes, they are described as being amorphous types.

Shape of the cells

- Different cells have different shapes. The cell wall in bacteria, algae, fungi and plant cells give shapes to the cells.
- In animal cells which lack cell wall, the cell membrane provides the boundary.
- The cells of bacteria (prokaryotes) have different shapes and they are classified based on the shapes as: Bacilli (rod shaped), Cocci (spherical shaped), Vibrio (comma shaped), Spirilla (spiral shape/helical shaped).
- Eukaryotes on the other hand (plants, animals, fungal cells) have the shape depending on the function;
- Human cells have various shapes: circular and biconcave (RBC), amoeboid (WBC), long and stretched and branched cells (nerve cells), long and narrow cells (columnar epithelium cells)
- Plant cells usually have square or rectangle shapes. Just as animal cells, some specialized cells have different shapes eg. Guard cells of stomata are kidney/dumbbell shaped.

- Fungal cells have a variety of shapes where they can be spherical, tubular or long chains.



Shapes of different cells

LECTURE 4: CELL ORGANELLES

Cell Membranes

- Both plant and animal cells are surrounded by an outer membrane referred to as the plasma membrane or the universal membrane.
- Most cell membranes are quite similar in structure though their properties vary greatly.
- Regardless of their origin, cell membranes contain proteins and lipids but many contain carbohydrates, though not all.
- Chemical analysis of cell membranes reveal that lipids are present and one type of lipid, the phospholipid, is prevalent.
- Singer and Nicolson, 1972 suggested a fluid mosaic model that will be used in the study of the structure of the cell membrane.

FUNCTIONS AND STRUCTURE OF A CELL MEMBRANE

(Use your own knowledge)

Cell Organelles

- Animal and plant organelles are enclosed by the membranes and are found suspended in a liquid matrix. Virtually all cells have nuclei, Eukaryotic cells without cells nuclei have short life spans and limited metabolic activities.
- Some cells have a nucleus at some time during their life cycle and lose it by the time they are fully differentiated. Cells contain other cells such as endoplasmic reticulum, mitochondria, vacuoles, Lysosome, and Ribosomes will be discussed. Some organelles are only present in one plant cells and absent in animal cells and vice versa.

LECTURE 5

MOVEMENT AND TRANSPORT

- Basically, movement is one feature that is used to determine whether an individual organism is alive or not.

- Some form of movement is the most obvious characteristic of living organisms.
- Movement is a response to a variety of stimuli and is a survival mechanism that enables organisms to adapt to the prevailing environmental condition and adopt to changes that may occur.
- Some kind of support is necessary and involved in movement of the whole or part of an organism.
- Most animals and plants are capable of locomotion that is moving their bodies as a whole from one place to another.
- Some of these organisms are sessile, that is attached to one place and only moving parts of their bodies.
- All cells are separated from their surroundings by the plasma membrane.
- Eukaryotic cells are further divided internally by a variety of membranes.
- Through these membranes a cell is capable of regulating the amount, kind, and often the direction of movement of substances that pass across the membranes.
- Water is the most important of the substances moving into and out of cells.

Movement in Plants

The type of movement differs with the type of stimuli, including:

- Nastic movement
- Tropism

- I. Haptotropism (pulling movement)
- II. Movement can either be nastic or tropic.
- III. Nastic movements in plants are reversible and repeatable movements in response to a stimulus whose direction is determined by the anatomy of the plant. For example Touch movement. Sleeping and awake movement. (Daisy plant that closes at dusk and opens in the morning)
- IV. Tropism is where different parts of the plant respond to different stimuli such as light (phototropism), humidity and gravity.

Movement in cells

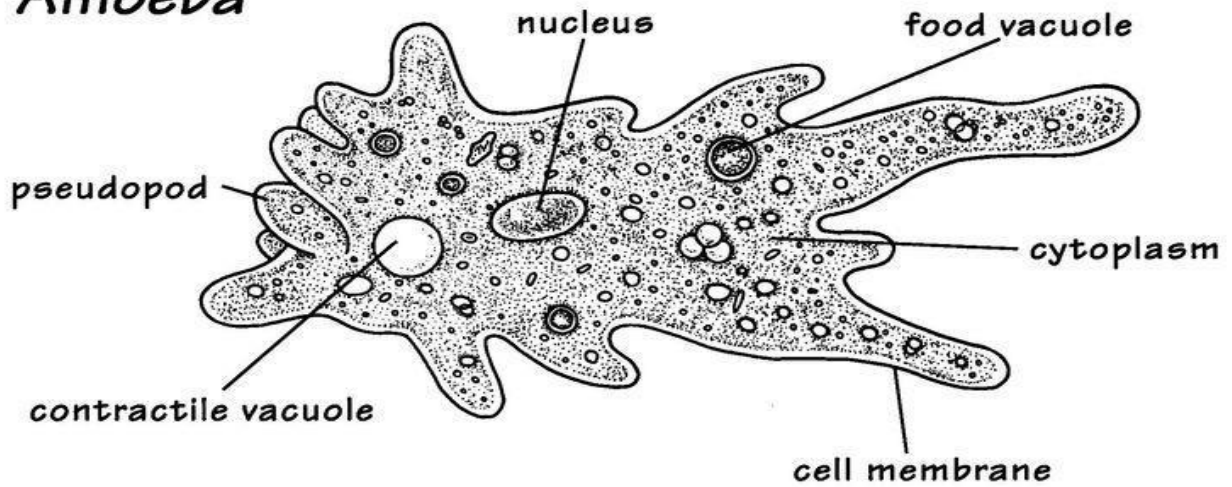
Living organisms display various types of movement.

These movement forms include;

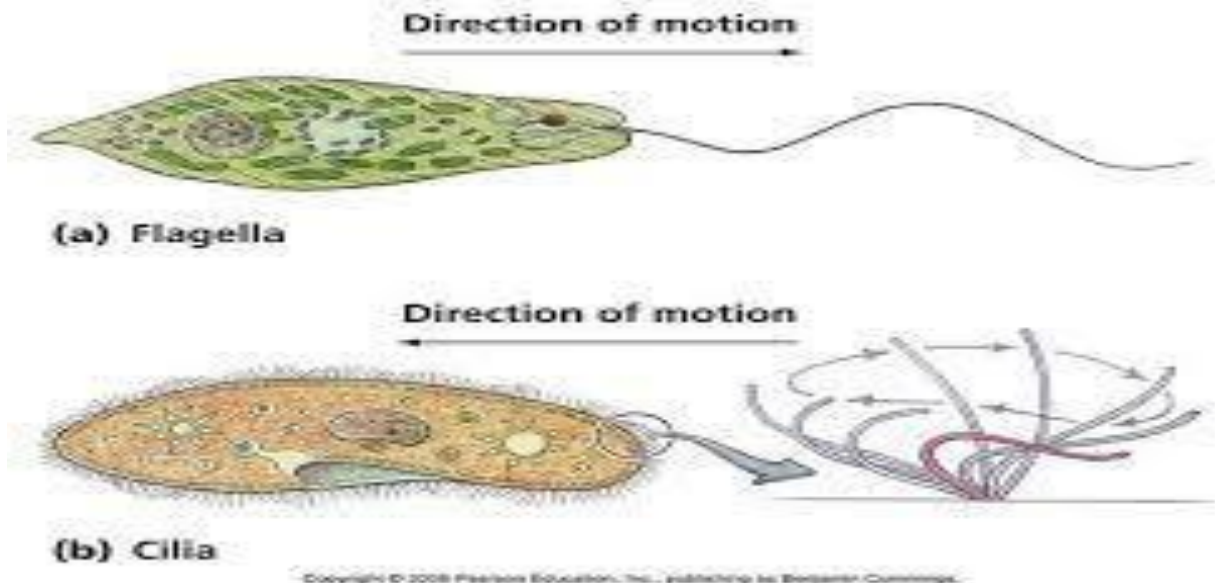
- Amoeboid movement,
- Ciliary and flagella motion,
- Every form of movement is aided by several features, for instance, amoeboid movement results from streaming of cytoplasm into advancing extensions of the cell surface, the pseudopods. This movement is exhibited not only by amoebae but also by various metazoan cells.

For example some human leucocytes move through body tissues and engulf bacteria, and embryonic cells migrate to new regions of the embryo during morphogenesis.

Amoeba



- Ciliary movement occurs in the most of our internal tubular organs which are lined with epithelium.
- The coordinated movement of cilia in the trachea help us in removing dust particles and some foreign substances inhaled along the atmospheric air.
- There is also movement of the ova through the female reproductive tract.
- Paramecium also moves in water using cilia.
- Flagella movement is similar to ciliary movement and can be seen in ciliary movement.



- Muscular movements: in vertebrates locomotion depends on the bones or endoskeleton and muscles attached to them.
- These muscles are called skeletal muscles. The bones and skeletal muscles together form musculoskeletal system.
- The movements of the body part are under voluntary control of the central nervous system. Thus they are also called voluntary muscles.

Movement of substances into and out of cells

- Surface areas of cells range from a minimum of 10 microns to 1,000 microns. The distance from any point in the environment to any point in a cell is quite small, the longest is 500 microns.
- The complexity of any transport system is a direct function of the distances involved.

-

Substances move across membranes through diffusion, osmosis, bulk flow of water, transport proteins and mass movement that occurs between cells.

- Transport that does not involve energy is said to be passive. Transportation using transport proteins such as the ATP and electrochemical gradient energy.

- **Diffusion**

- Oxygen will diffuse from a high concentration at the alveoli in the lungs, to a lower concentration in the red blood cells.
- This is a passive process, which means it does not need energy.
- The oxygen diffuses across the partially permeable membrane of the alveoli cells and the capillary wall, into the red blood cells to be carried around the body.
- **Osmosis** is the net movement of water molecules from a high concentration to a lower concentration across a partially permeable membrane.
- Osmosis is a passive process so does not require energy.

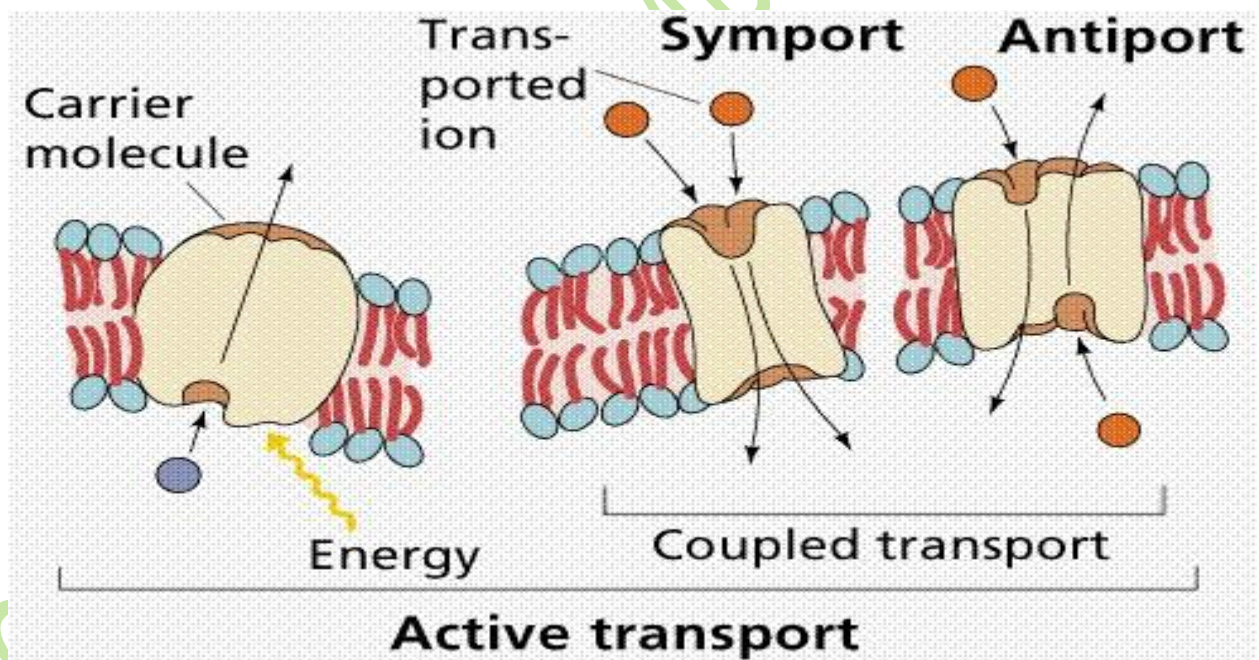
- **Active transport**

- Some substances that need to get into cells get there by **active transport**.

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- This transport process moves substances against the concentration gradient - the substances move from a low concentration to a higher concentration.
- It requires an energy input.

Glucose can be taken into cells by active transport if the concentration of glucose inside the cell is already quite high.

- This requires **carrier proteins**.



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LESSON 6

MACROMOLECULES OF LIFE-NUCLEIC

ACIDS AND PROTEINS

Introduction

- Cells are composed of many substances of various complexity.
- Some substances are found as atoms, others as molecules, small monomers, complex monomers and polymers.

Nucleic acids, DNA and RNA are now known as the genetic materials, DNA being the primary genetic material. The genetic material an organism inherits from its parent make it look, be what they are in terms of various characteristics or traits and behave or act the way they behave or act.

DEFINITION

- DNA or rather Deoxyribonucleic acid is a molecule that contains the instructions an organism needs to develop, live and reproduce.
- These instructions are found inside every cell, and are passed down from parents to children.

Deoxyribonucleic Acid structure

- DNA composed of complex monomers/molecules called nucleotides.

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- A DNA molecule is a complex biological polymer made up of repeated nucleotides (poly- nucleotides).
- The nucleotides are made up of a Nitrogen base (Adenine, Guanine, Cytosine, Thiamine), phosphate group and sugars.
- The order of these basses is what determines DNA's instructions, or genetic code.
- Human DNA has around 3 billion bases, and more than 99 percent of those bases are the same in all people in the world.

Just as the words of the alphabet form words, the order of nitrogen bases in a DNA sequence form genes.

- These are the deemed as the language of the cell since they are the ones that direct the cells in making proteins.
- The RNA is the one that translates the genetic information from DNA into proteins.
- The nucleotides are joined together by phosphodiester bond.
- Nucleotides are attached together to form two long strands that spiral to create a structure called a double helix.
- Two polypeptides make up the double helix by winding around each other. Bases that are one-ringed makes up the pyrimidines (Thymine and Cytosine) while two ringed bases make purines (adenine and guanine).

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- After every 10 base pairs the helix makes a complete turn (360°). The distance between two base pairs is 0.34 nm.
- This means that each base pair is twisted 360° clockwise with respect to the previous base pair.
- Because of the way bases bond with each other, the sugarphosphate backbone is not equally spaced along the helical axis.
- This results to formation of minor and major grooves
- DNA molecules are long-for them to fit in the cells, it is tightly coiled to form structures (chromosomes).
- Where each chromosome contains a single DNA molecule.
- Humans have 23 pairs of chromosomes, found inside the cells.

Biological function of DNA include:

1. The DNA polymers direct the production of other polymers called proteins.
2. A chromosome consists of smaller segments called genes.

Structure and Functions of Ribonucleic Acid

- Just like DNA, RNA is a long chain of macromolecules of nucleic acid but it differs from DNA in the following important properties:

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- In the nucleotides of RNA the sugar component is ribose and not deoxyribose.
- The nitrogenous base Thymine is found in DNA but does not occur in RNA. Instead it is replaced with Uracil in RNA.
- RNA is usually single stranded and does not form a regular helical structure.
- RNA is generally shorter than DNA.
- RNA can exist in various forms such as mRNA, tRNA or rRNA which have different structures and roles.

DNA AS A CODE FOR PROTEIN SYNTHESIS AND THE PRIMARY GENETIC MATERIAL • Gene expression

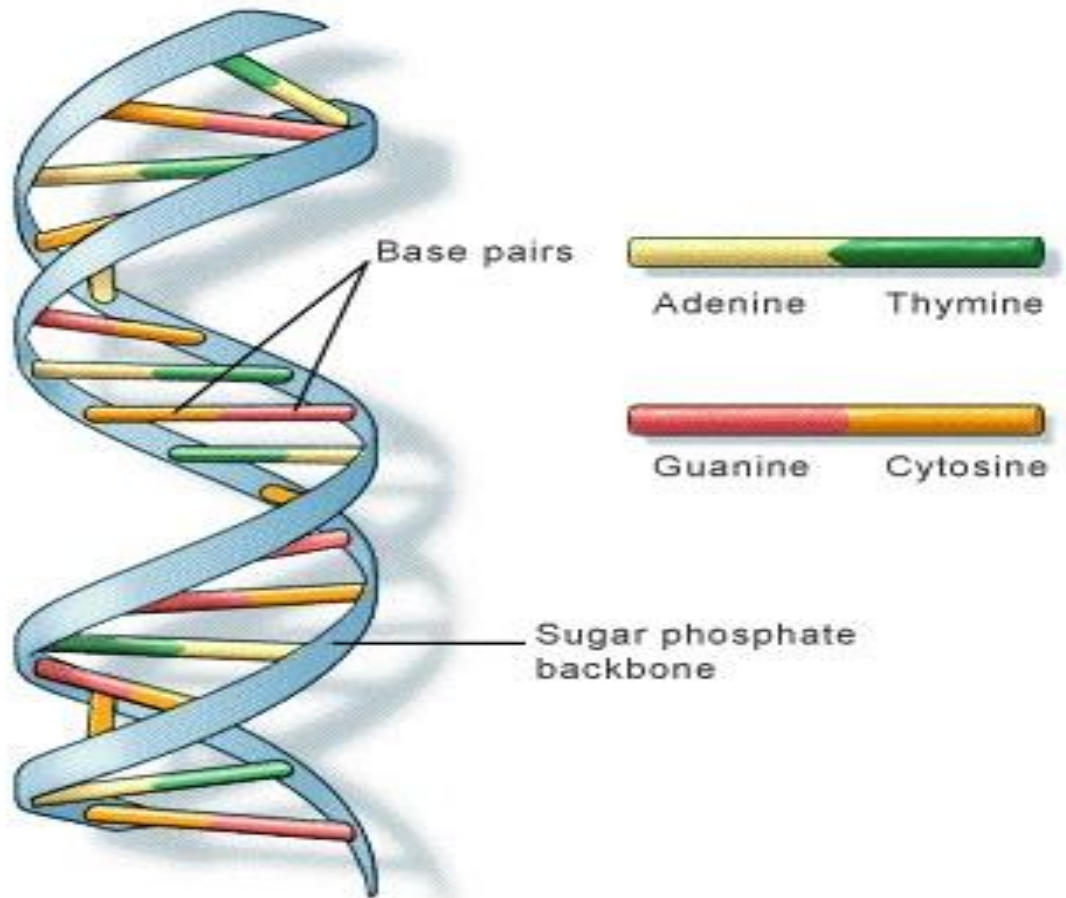
DNA is the genetic material of all organism on Earth. When DNA is transmitted from parents to children, it can determine some of the children's characteristics (such as their eye color or hair color).

• Genes specify functional products

- A DNA molecule is a long, string of nucleotides.
- It is divided up into functional units called **genes**.
- Each gene provides instructions for a functional product, that is, a molecule needed to perform a job in the cell.

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- In many cases, the functional product of a gene is a protein. For example, Mendel's flower color gene provides instructions for a protein that helps make colored molecules (pigments) in flower petals.

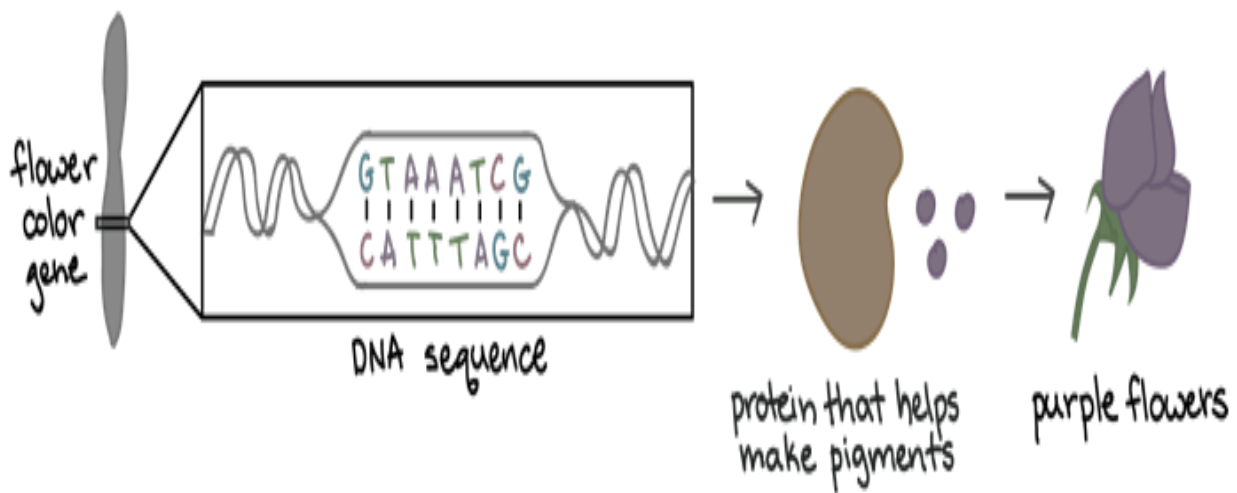
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- Diagram of how a gene can dictate a phenotype (observable feature) of an organism.
- The flower color gene that Mendel studied consists of a stretch of DNA found on a chromosome.
- The DNA has a particular sequence; part of it, shown in this diagram, is 5'-GTAAATCG-3' (upper strand), paired with the complementary sequence 3'-CATTTAGC-5' (lower strand).
- The DNA of the gene specifies production of a protein that helps make pigments.
- When the protein is present and functional, pigments are produced, and the flowers of a plant have a purple color.

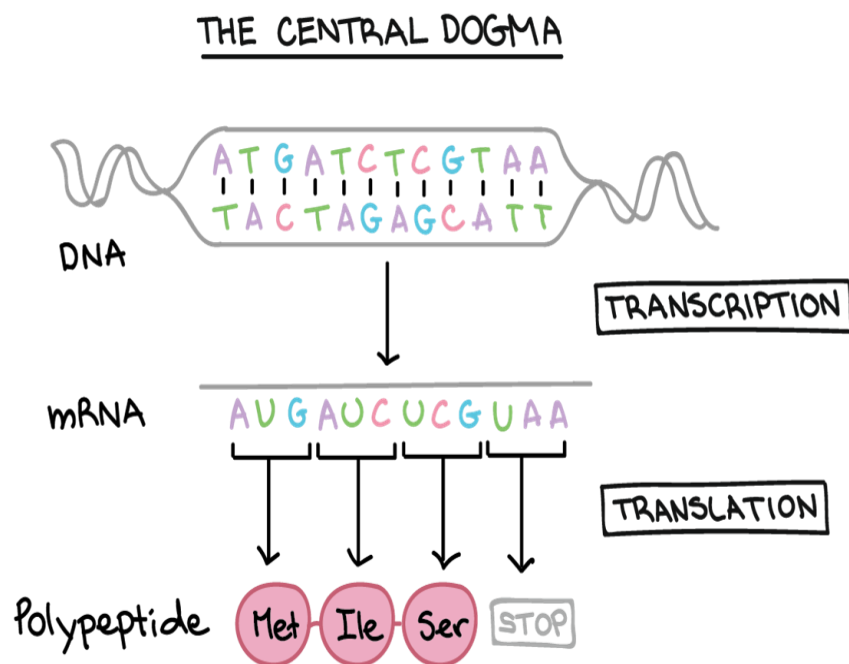
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- The functional products of most known genes are proteins, or, more accurately, polypeptides.
- Polypeptide is another word for a chain of amino acids. Although many proteins consist of a single polypeptide, some are made up of multiple polypeptides. Genes that specify polypeptides are called protein-coding genes.
- Not all genes specify polypeptides. Instead, some provide instructions to build functional RNA molecules, such as the [transfer RNAs](#) and [ribosomal RNAs](#) that play roles in translation.
- HOW DOES THE DNA SEQUENCE OF A GENE SPECIFY A PARTICULAR PROTEIN?
- Many genes provide instructions for building polypeptides. How, exactly, does DNA direct the construction of a polypeptide? This process involves two major steps: transcription and translation.
- In transcription, the DNA sequence of a gene is copied to make an

RNA molecule. This step is called *transcription* because it involves rewriting, or transcribing, the DNA sequence in a similar RNA "alphabet." In [eukaryotes](#), the RNA molecule must undergo processing to become a mature messenger RNA (mRNA).

- In **translation**, the sequence of the mRNA is decoded to specify the amino acid sequence of a polypeptide.
- The name *translation* reflects that the nucleotide sequence of the mRNA sequence must be translated into the completely different "language" of amino acids.
- **THE CENTRAL DOGMA**



The two strands of DNA have the following sequences:

5'-ATGATCTCGTAA-3'

3'-TACTAGAGCATT-5'

Transcription of one of the strands of DNA produces an mRNA that nearly matches the other strand of DNA in sequence. However, due to a biochemical difference between DNA and RNA, the Ts of DNA are replaced with Us in the mRNA. The mRNA sequence is:

5'-AUGAUCUCGUAA-5'

Translation involves reading the mRNA nucleotides in groups of three; each group specifies an amino acid (or provides a stop signal indicating that translation is finished).

3'-AUG AUC UCG UAA-5'

AUG → Methionine

AUC → Isoleucine

UCG → Serine

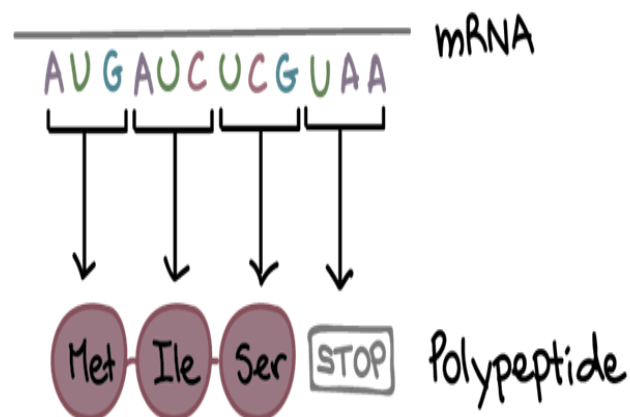
UAA "Stop"

Polypeptide sequence: (N-terminus) Methionine-Isoleucine-Serine (C-terminus)

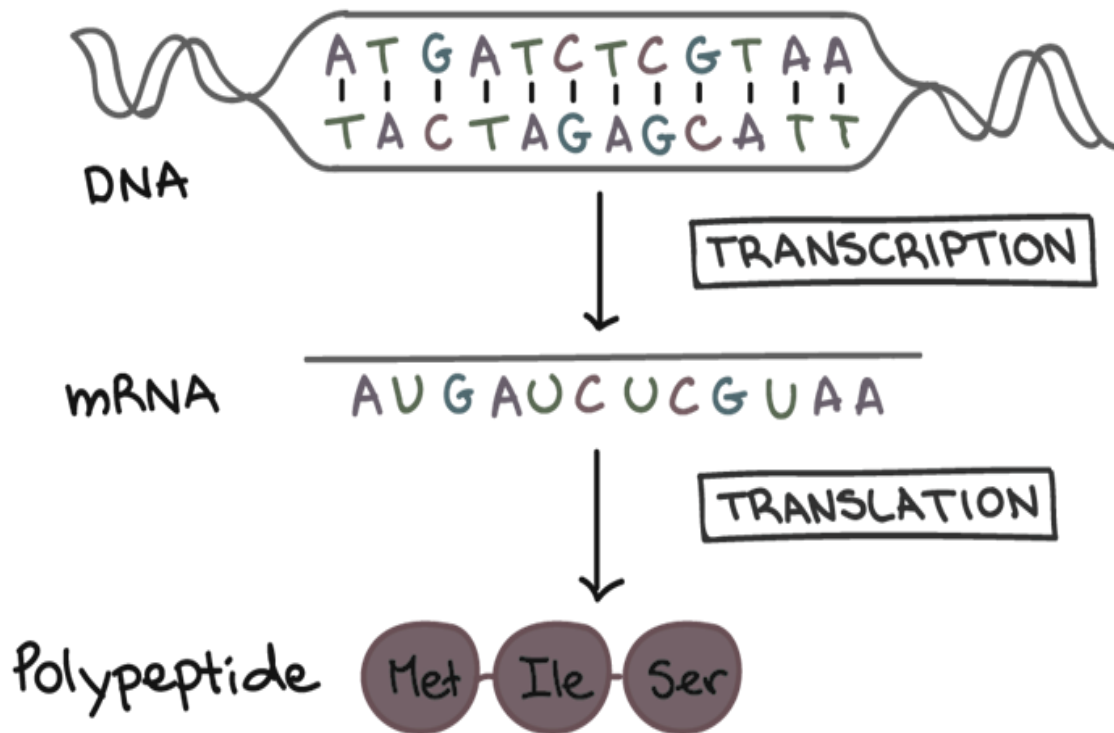
- Thus, during expression of a protein-coding gene, information flows from DNA → RNA → protein. This directional flow of information is known as the **central dogma** of molecular biology. Non-protein-coding genes (genes that specify functional RNAs) are still transcribed to produce an RNA, but this RNA is not translated into a polypeptide. For either type of gene, the process of going from DNA to a functional product is known as **gene expression**.

- **THE GENETIC CODE**

- During translation, the nucleotide sequence of an mRNA is translated into the amino acid sequence of a polypeptide. Specifically, the nucleotides of the mRNA are read in triplets (groups of three) called **codons**. There are 61 codons that specify amino acids.
- One codon is a "**start**" **codon** that indicates where to start translation. The start codon specifies the amino acid methionine, so most polypeptides begin with this amino acid. Three other "**stop**" **codons** signal the end of a polypeptide. These [relationships](#) between codons and amino acids are called the **genetic code**.



- **GENE EXPRESSION AND THE GENETIC CODE**



• PROPERTIES OF A GENETIC CODE

- The codons of an mRNA are “read” one by one inside protein- and RNA structures called [ribosomes](#), starting at the 5’ end of the gene and moving towards the 3’ end.
- Translation always begins at a **start codon**, which has the sequence AUG and encodes the amino acid methionine (Met) in most organisms. Thus, every polypeptide typically starts with methionine, although the initial methionine may be snipped off in later processing steps. A start codon is required to begin translation, but the codon AUG can also appear later in the coding sequence of an mRNA, where it simply specifies the amino acid methionine.

		Second letter					
		U	C	A	G		
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G	Third letter
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G	
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G	
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G	

- Genetic code table: Each letter sequence of mRNA nucleotides corresponds to a specific amino acid, or to a stop codon. UGA, UAA and UAG are stop codons while AUG is a start codon
- Once translation has begun at the start codon, the following codons of the mRNA will be read one by one, in the 5' to 3' direction. As each codon is read, the matching amino acid is added to the C-terminus of the polypeptide. Most of the codons in the genetic code specify amino acids and are read during this phase of translation.
- Translation continues until a **stop codon** is reached. There are three stop codons in the genetic code, UAA, UAG, and UGA. Unlike start codons, stop codons don't correspond to an amino acid.

Instead, they act as "stop" signals, indicating that the polypeptide is complete and causing it to be released from the ribosome. More nucleotides may appear after the stop codon in the mRNA, but will not be translated as part of the polypeptide.

. Degenerate code

- As previously mentioned, the genetic code consists of 64 unique codons. But if there are only 20 amino acids, what are the other 44 codons doing? As we saw, a few are stop codons, but most are not. Instead, the genetic code turns out to be a **degenerate code**, meaning that some amino acids are specified by more than one codon. For example, proline is represented by four different codons (CCU, CCC, CCA, and CCG). If any one of these codons appears in an mRNA, it will cause proline to be added to the polypeptide chain.
- Most of the amino acids in the genetic code are encoded by at least two codons. In fact, methionine and tryptophan are the only amino acids specified by a single codon. Importantly, the reverse isn't true: each codon specifies just one amino acid or stop signal

2. The genetic code is unambiguous:

- There is no uncertainty in the genetic code. A particular codon in mRNA will always be predictably translated into a particular amino acid or stop signal. the start signal will always be AUG.

3. The genetic code is universal

- With some minor exceptions, all living organisms on Earth use the same genetic code. This means that the codons specifying

the 20 amino acids in your cells are the same as those used by the bacteria inhabiting hydrothermal vents at the bottom of the Pacific Ocean. Even in organisms that don't use the "standard" code, the differences are relatively small, such as a change in the amino acid encoded by a particular codon.

- A genetic code shared by diverse organisms provides important evidence for the common origin of life on Earth. That is, the many species on Earth today likely evolved from an ancestral organism in which the genetic code was already present

4. The genetic code is non-overlapping.

- The reading of the code or replication is in one direction in the same process and no skipping a codon or process.
- However viruses do not use the same system, they have a few minor changes to the genetic coding making them different.

LESSON 7

PROTEIN BIOSYNTHESIS

Introduction

- Protein synthesis is also referred to as translation. This is because it involves the transfer of information from one language (sequence of nucleotides) to another (sequence of amino-acids).

- This process involves three main stages; Initiation, elongation of the polypeptide and termination. The process starts at the

initiator codon mainly the AUG and rarely at the GUG codon of the mRNAs. Translation uses the products the products of transcription as had been discussed in the previous lesson.

- This lesson will focus on the process through which the information coded in the DNA is decoded to form proteins.

Transcription

- It takes place in the nucleus
- This is the process of production of messenger RNA (mRNA) by the enzyme RNA polymerase and the processing of the resulting mRNA molecule.
- It uses DNA as a template to make mRNA molecule
- During transcription, a strand of mRNA is made that is complementary to a strand of DNA.
- It involves four steps: Initiation, Elongation, Termination and processing

Initiation

It is the beginning of transcription. It occurs when the enzyme RNA polymerase binds to a region of a gene called the promoter.

This signals the DNA to unwind so the enzyme can “read” the bases in one of the DNA strands.

The enzyme is ready to make a strand of mRNA with a complementary sequence of bases .

Initiation

- It is the beginning of transcription.

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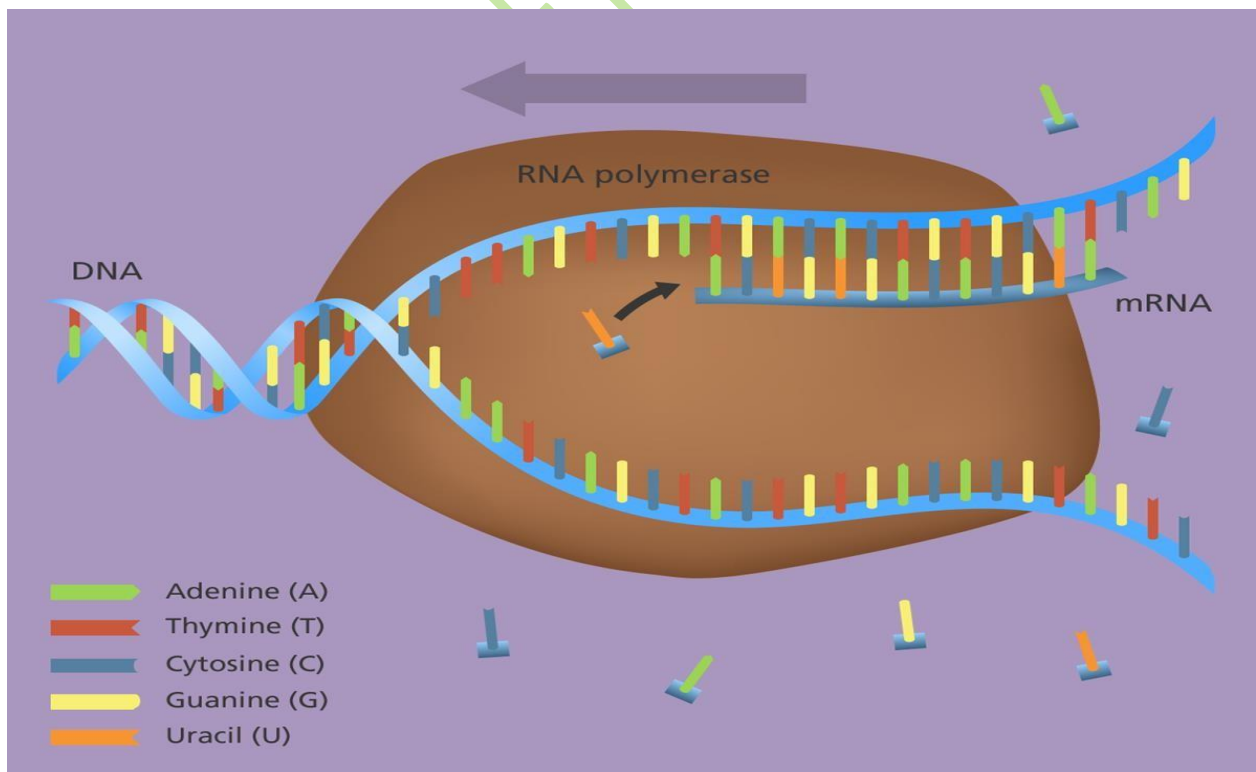
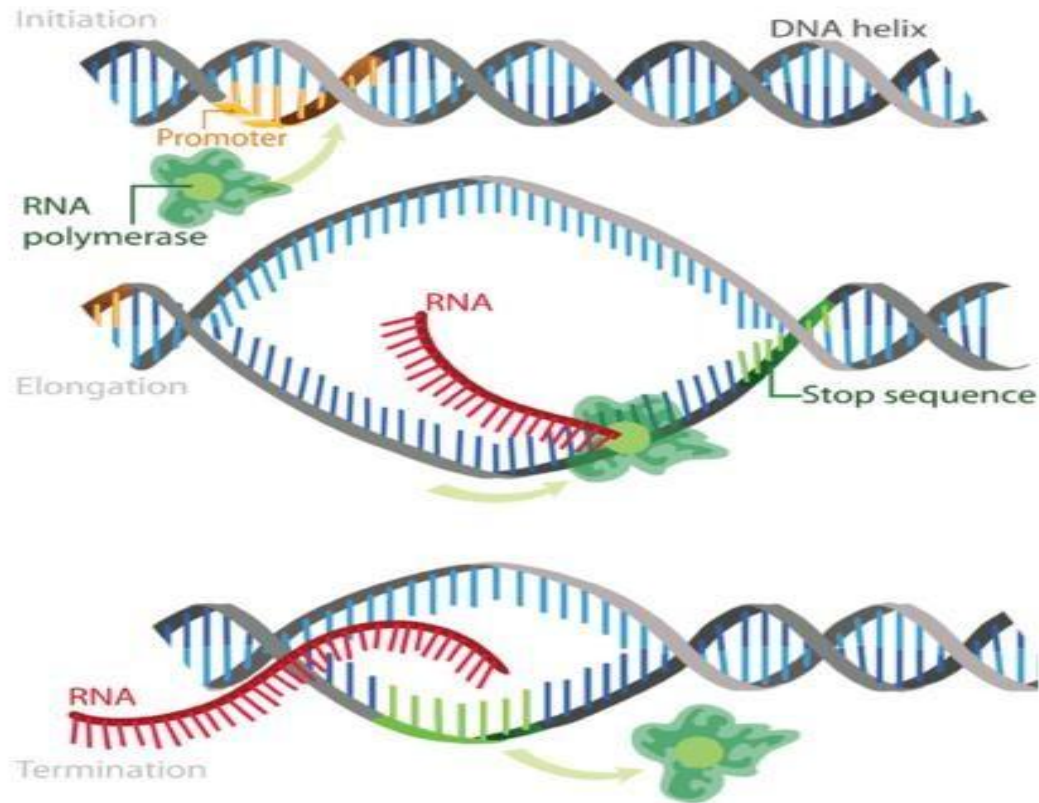
- It occurs when the enzyme RNA polymerase binds to a region of a gene called the promoter.
- This signals the DNA to unwind so the enzyme can “read” the bases in one of the DNA strands.

Termination

- This is the ending of transcription.
- It occurs when RNA polymerase crosses a stop (termination) sequence in the gene.
- The mRNA is complete and detaches from DNA

Processing

- After transcription, the RNA molecule is processed in a number of ways:
- Introns are removed
- Exons are spliced together to form a mature mRNA molecule consisting of a single protein- coding sequence



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Translation

- It takes place in the ribosome (large and small subunit)
- This is the use of mRNA to direct protein synthesis and the subsequent post translational processing of the protein molecule.
- The mature mRNA molecule is used as a template to assemble a series of amino acids to produce a polypeptide with specific amino acid sequence.
- It takes place in four stages;
- Initiation
- Elongation
- Termination
- Posttranslational modification of protein

Initiation

- The small subunit of the ribosome binds at the 5' end of the mRNA molecule and moves in a 3' direction until it meets a start codon (AUG)
- It then forms a complex with the large unit of the ribosome complex and an initiation tRNA molecule.

Elongation

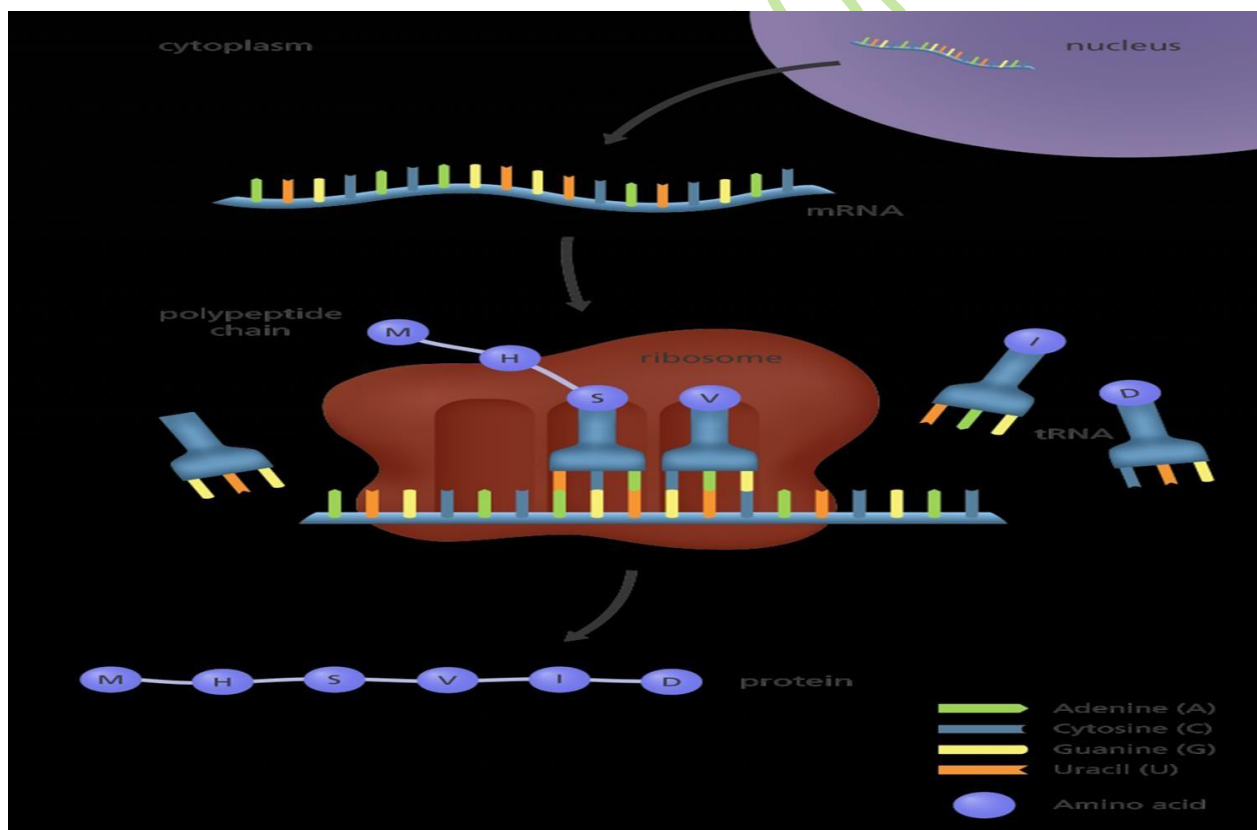
- The subsequent codons on the mRNA molecule determine which tRNA molecule linked to an amino acid binds to the mRNA.
- An enzyme peptidyl transferase links the aminoacids together using peptide bonds.

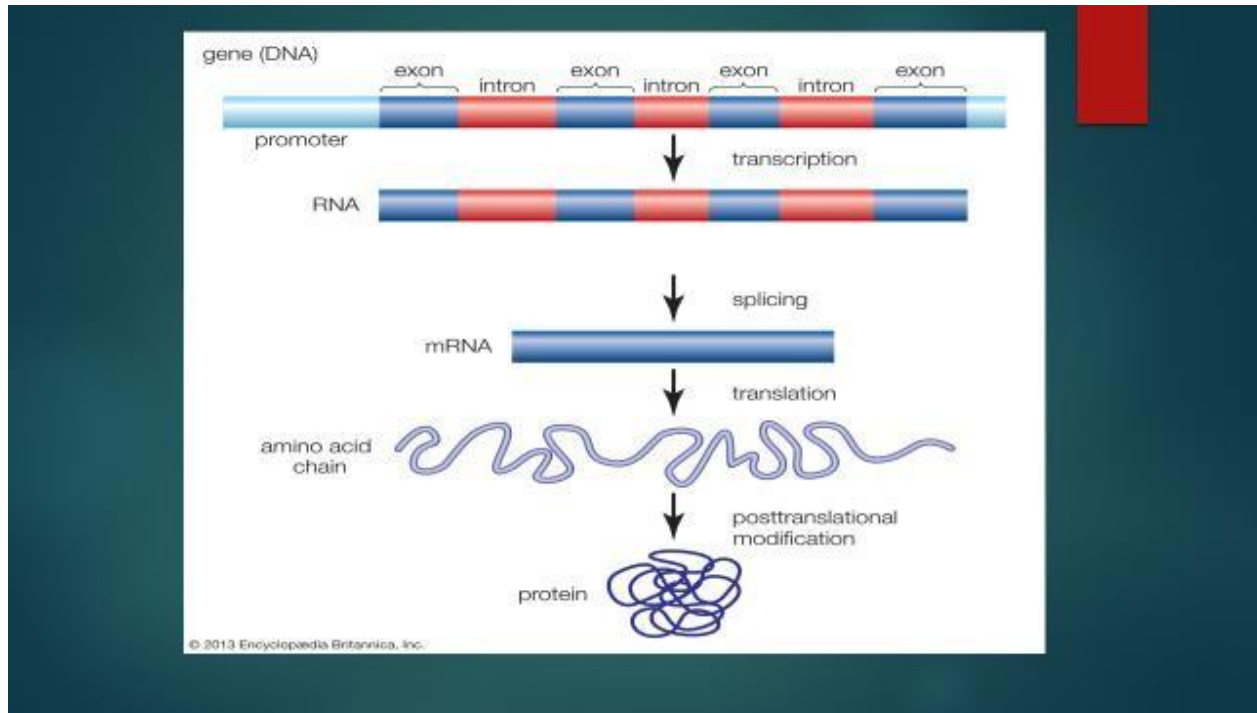
- The process continues, producing a chain of amino acids as the ribosome moves along the mRNA molecule.

Termination

- It occurs when the ribosome encounters a stop codon.
- The stop codon are: UAA, UAG, or UGA

The small and large ribosomal subunits separate from the mRNA and from each other, each element can now participate in another round of translation





Posttranslational modification of protein (PTMs)

- These are chemical modifications of a polypeptide chain that occur after DNA has been transcribed into RNA and translated into protein.
- They range from:
- the enzymatic cleavage of peptide bonds
- covalent additions of particular chemical groups, lipids, carbohydrates, or even entire proteins to amino acid side chains.
- They diversifies structures and functions of proteins.
- Posttranslational modifications (PTMs) of proteins such as phosphorylation and ubiquitination are crucial for controlling protein stability, localization, and conformation.

Translation

- In both prokaryotes and eukaryotes, translation involves three main stages:
 - i. Initiation of translation
 - ii. Elongation of polypeptide chain
 - iii. Termination of translation
- iv. **Initiation of translation**
 - v. In prokaryotes, the mRNA is not processed so it is modified by addition of formyl group to form formylmethionine (fmet). Enzyme transformalase catalyses addition of formyl group to form a fmet_tRNA (fmet).
- 1) Prokaryotic mRNA has a purine rich region that binds to a pyrimidine rich region of the 16S rRNA. Eukaryotes do not have the Shine-Dalgarno sequence. They recognise the cap at the 5' end of mRNA.

ii) Elongation of polypeptide chain

- After initiation, a second charged tRNA moves in the site A- site while fmet remains at the P- site.
- The ribosome puts the second tRNA in the correct position so that a peptide bond is formed between the two amino acids.
- This reaction is catalysed by enzyme peptidyl transferase. The ribosome moves along the mRNA towards the 3' end in a process called translocation.

- The ribosome moves along the mRNA towards the 3' end in a process called translocation. When several ribosomes are translating a single mRNA, it is referred to as a polysome or polyribosome.

iii) Termination of translation

- The end of a polypeptide chain is signalled by one of the three stop codons (AUG, UAA or UGA). Ribosomes recognise stop codons by help of proteins called termination factors or release factors

Assignment

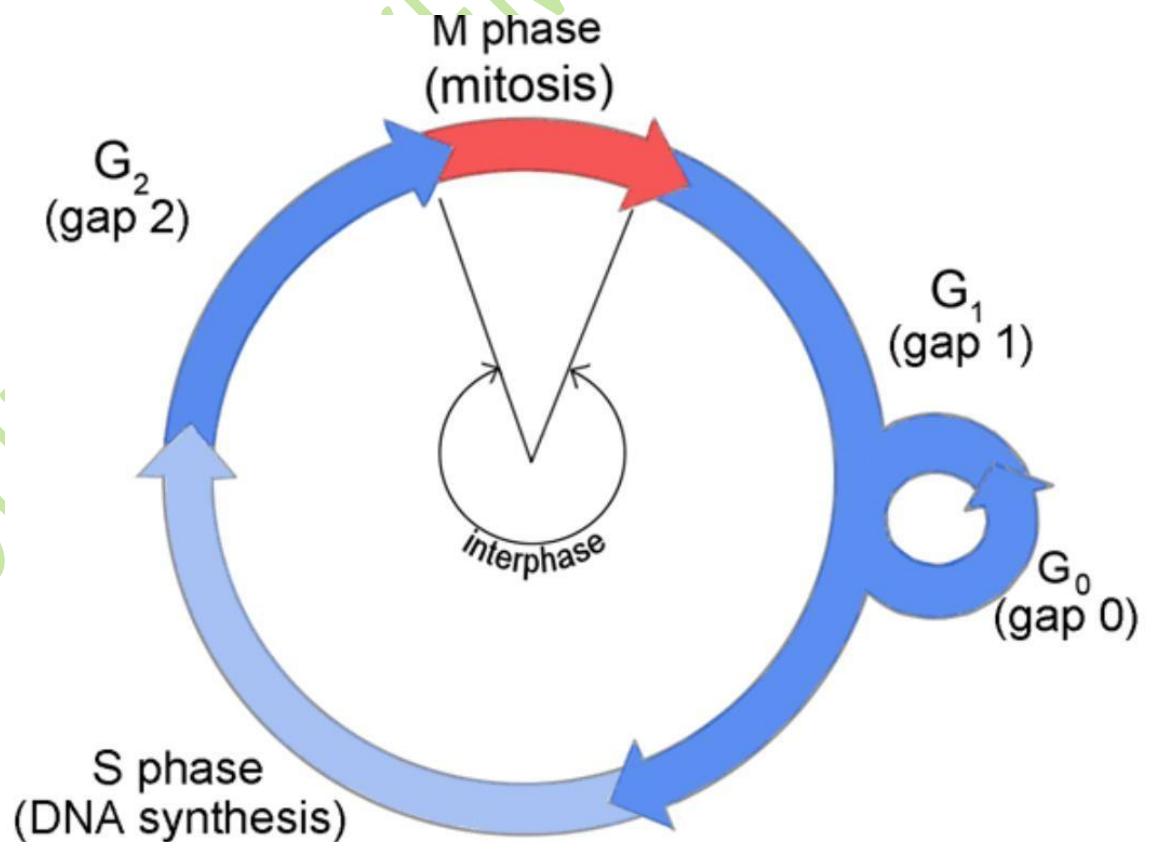
- Discuss the roles of DNA, mRNA, tRNA and rRNA in protein synthesis

PHYSICAL BASIS OF HEREDITY

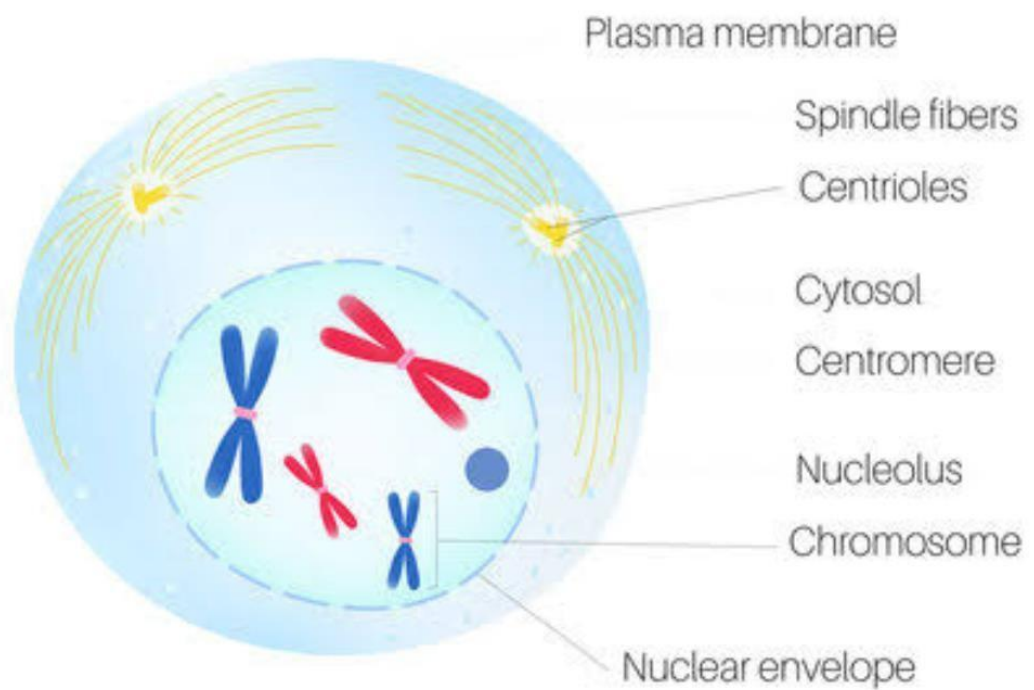
- Since time in memorial even before heredity material was identified, its been discovered that chromosomes contained genetic material that passed characteristics from parents to their offsprings.
- This was revealed from chromosome behavior during cell divisions in mitosis and meiosis. These are the processes that lead to sharing and passing of hereditary characteristics from parent to daughter cells.

Mitosis Cell Division

- When a cell starts to undergo division by mitosis each of its chromosomes consists of a pair of identical sister chromatids, joined at the centromeres.
- During mitosis each of the daughter cells receives an identical number of chromosomes with a complete set of genetic information.
- This process is important since it is the major method of asexual reproduction in both plants and some animals. It takes place in five steps, Interphase, prophase, metaphase, anaphase and telophase.



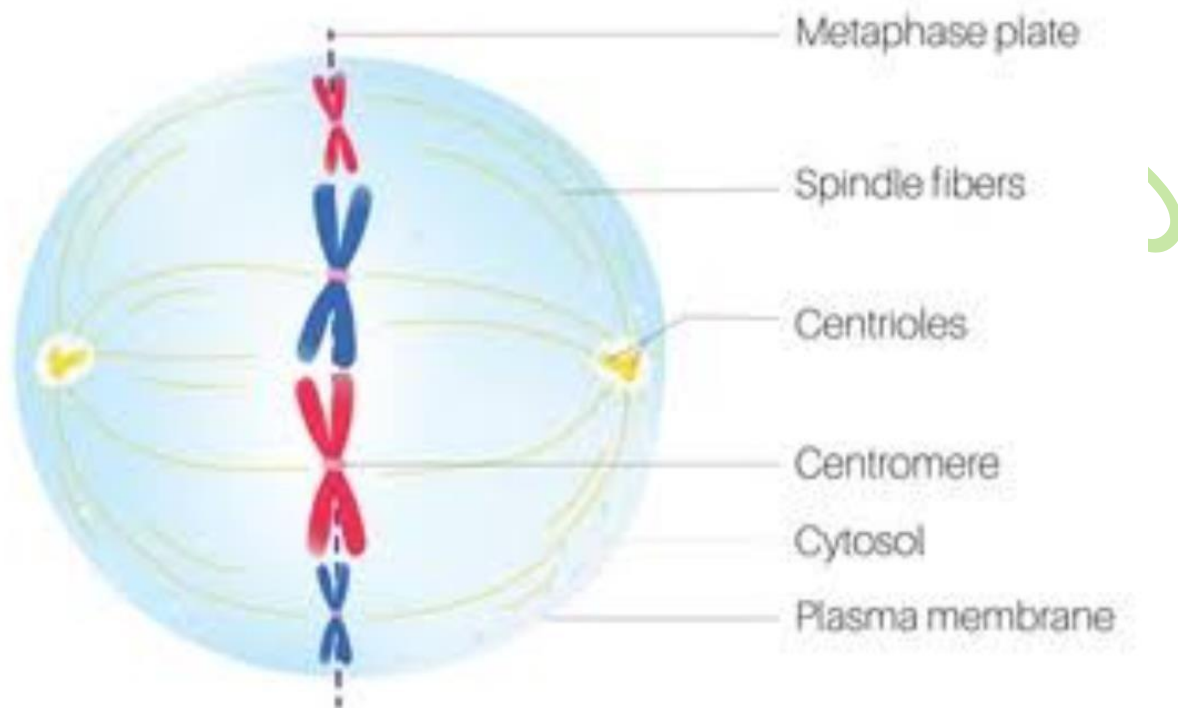
PROPHASE



Prophase stage

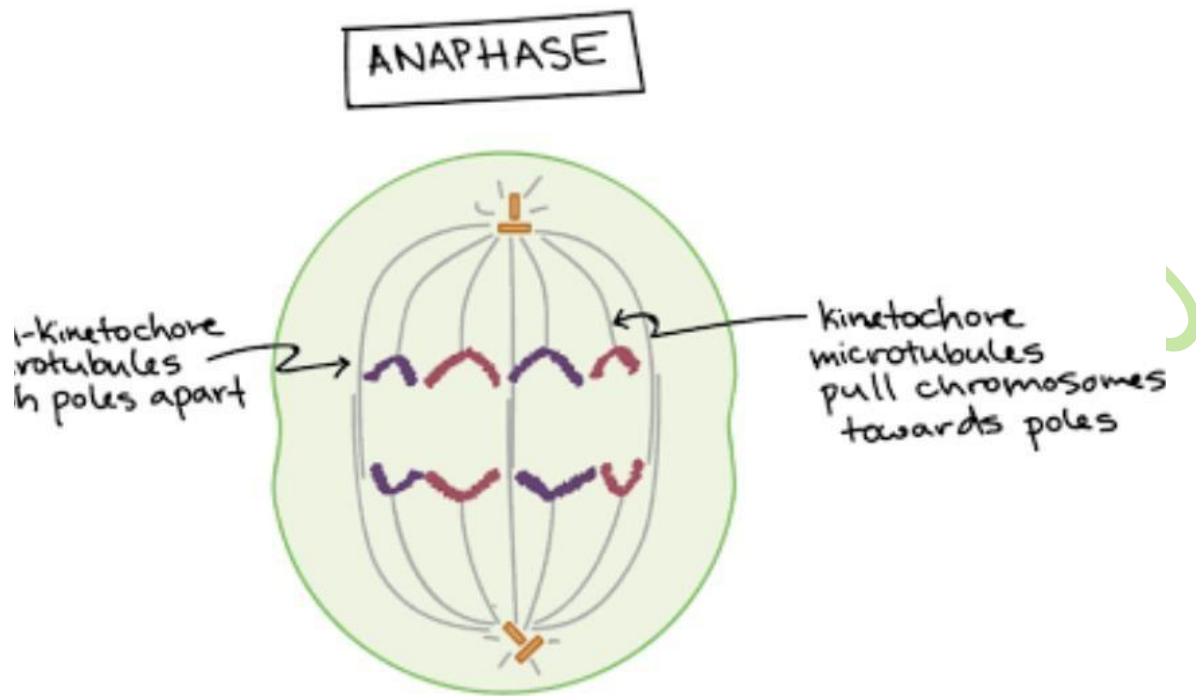
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METAPHASE



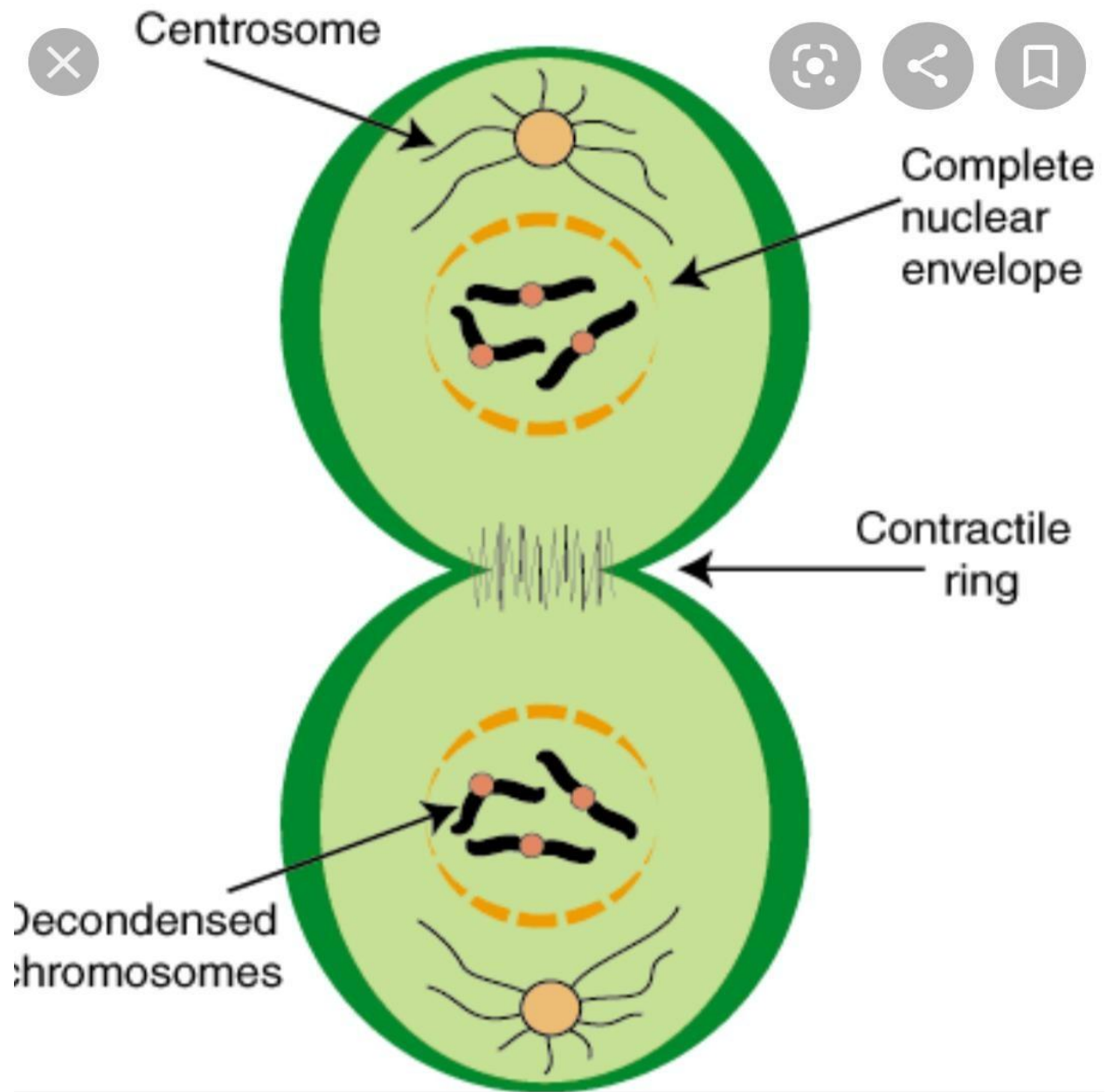
Metaphase stage

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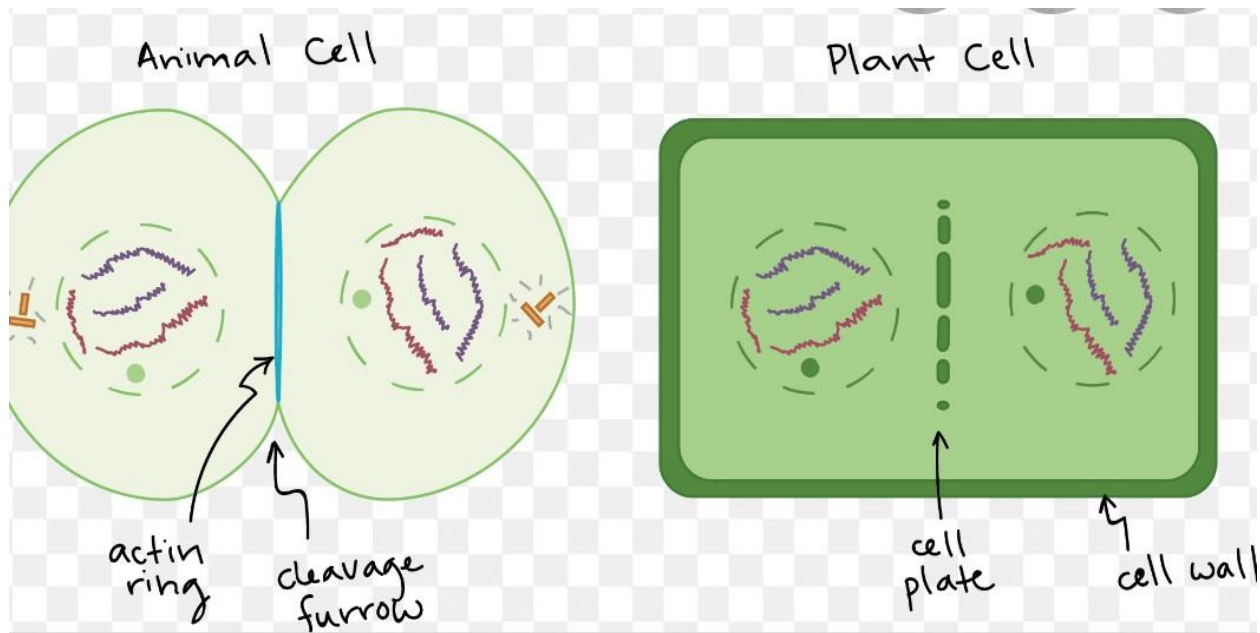
Anaphase Stage

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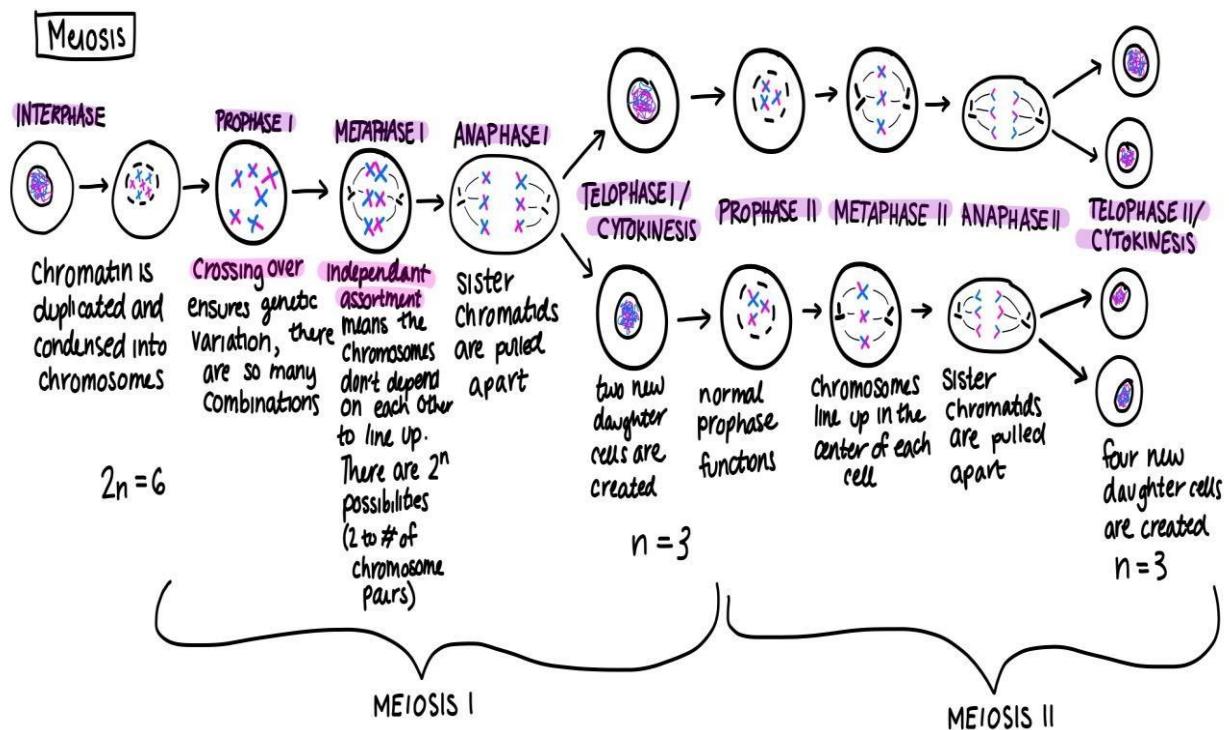
Anaphase Stage



Phases of meiosis

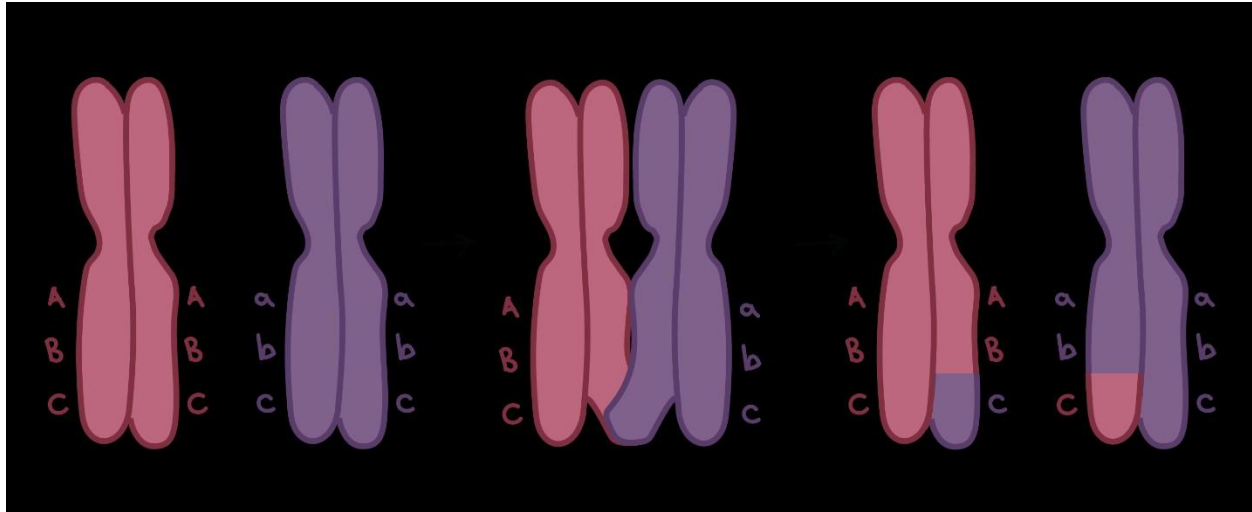
- In many ways, meiosis is a lot like mitosis. The cell goes through similar stages and uses similar strategies to organize and separate chromosomes.
- In meiosis, however, the cell has a more complex task. It still needs to separate sister chromatids (the two halves of a duplicated chromosome), as in mitosis.

- But it must also separate homologous chromosomes, the similar but nonidentical chromosome pairs an organism receives from its two parents.
- These goals are accomplished in meiosis using a two-step division process. Homologue pairs separate during a first round of cell division, called meiosis I. Sister chromatids separate during a second round, called meiosis II.
- Since cell division occurs twice during meiosis, one starting cell can produce four gametes (eggs or sperm). In each round of division, cells go through four stages: prophase, metaphase, anaphase, and telophase.



Meiosis I

- Before entering meiosis I, a cell must first go through interphase. As in mitosis, the cell grows during G1, copies all of its chromosomes during S phase, and prepares for division during G2.
- During **prophase I**, differences from mitosis begin to appear. As in mitosis, the chromosomes begin to condense, but in meiosis I, they also pair up. Each chromosome carefully aligns with its homologue partner so that the two match up at corresponding positions along their full length.
- For instance, in the image in the next slide, the letters A, B, and C represent genes found at particular spots on the chromosome, with capital and lowercase letters for different forms, or alleles, of each gene. The DNA is broken at the same spot on each homologue—here, between genes B and C—and reconnected in a criss-cross pattern so that the homologues exchange part of their DNA.
- This process, in which homologous chromosomes trade parts, is called **crossing over**. It's helped along by a protein structure called the **synaptonemal complex** that holds the homologues together. The chromosomes would actually be positioned one on top of the other throughout crossing over; they're only shown side-by-side in the image above so that it's easier to see the exchange of genetic material.



- In **anaphase I**, the homologues are pulled apart and move apart to opposite ends of the cell. The sister chromatids of each chromosome, however, remain attached to one another and don't come apart.
- Finally, in **telophase I**, the chromosomes arrive at opposite poles of the cell. In some organisms, the nuclear membrane re-forms and the chromosomes decondense, although in others, this step is skipped—since cells will soon go through another round of division, meiosis. Cytokinesis usually occurs at the same time as telophase I, forming two haploid daughter cells.

Meiosis II

- Cells move from meiosis I to meiosis II without copying their DNA. Meiosis II is a shorter and simpler process than meiosis I, and you may find it helpful to think of meiosis II as “mitosis for haploid cells.”

- The cells that enter meiosis II are the ones made in meiosis I. These cells are haploid—have just one chromosome from each homologue pair—but their chromosomes still consist of two sister chromatids. In meiosis II, the sister chromatids separate, making haploid cells with non-duplicated chromosomes.

Meiosis I

Early Prophase I:
 -Chromosomes and centrioles duplicate
 -Centrioles begin to move

Late Prophase I:
 -Chromosomes line up in homologous pairs
 -Spindle begins to form

Metaphase I:
 -Chromosomes line up on metaphase plate
 -Microtubules attach to one homologue each

Anaphase I:
 -Microtubules pull and separate pairs of chromosomes

Telophase I:
 -Cell membrane constricts
 -Chromosomes are surrounded by nuclear material

End of Meiosis I:
 -Two haploid cells
 -Chromosomes are still duplicated, so Meiosis II begins

Metaphase II:
 -Chromosomes from each cell line up on metaphase plate

Prophase II:
 -Both cells duplicate centrioles and form spindles

Anaphase II:
 -Microtubules pull the duplicate chromosomes from originals

Telophase II:
 -Cell membranes constrict
 -Nuclei form

End of Meiosis II:
 -Four haploid cells (gametes), with no duplicated chromosomes

Meiosis II



Success comrade